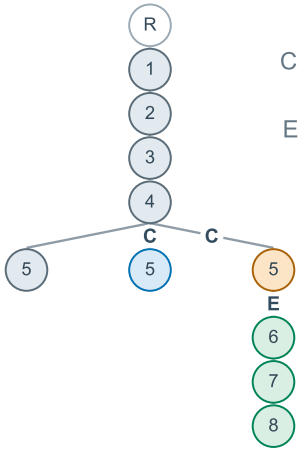
Let S be the set of positive integer divisors of 10^9. Compute the number of subsets T of S such that
- for every element s of S, exactly one of s and 10^9/s is in T, and
- for every element t of T, all positive integer divisors of t are in T.

Extracted CoT

10^9=2^9 5^9 divisors grid 10x10, ideals self complementary under rotation. Count symmetric monotonic paths length20 from (0,10) to (10,0), symmetric under 180 => first10 arbitrary 5? First half endpoints must center (5,5) for rotational path invariant since middle vertex center. Thus first half 5 each gives binom10,5=252. Verify small n=2 grid ideals self comp 2 yes binom2,1; n=4 6 yes. Could give heights h0>=...>=h9 between0,10 and h_i+h9-i=10. first5 decreasing h0..h4>=5 <=10, latter fixed, so multiset 5 from6 => C10,5=252. Explain height conditions including central h4>=h5=10-h4 => h4>=5. first5 any weak dec in5..10. exact 252.

Reasoning Tree



R = Root
C = Continuous Logic
E = Exploration